

Plain Text

```
"protein": 0, "carbs": 0, "fat": 0}` | Macronutrient data |
```

```
*  **SQL for `fitness_data` table:**
  ```sql
CREATE TABLE fitness_data (
 id BIGINT GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
 user_id UUID REFERENCES users(user_id) ON DELETE CASCADE,
 timestamp TIMESTAMPTZ DEFAULT now(),
 steps INTEGER DEFAULT 0,
 calories_burned INTEGER DEFAULT 0,
 heart_rate INTEGER DEFAULT 0,
 macros JSONB DEFAULT '{"protein": 0, "carbs": 0, "fat": 0}':::jsonb
);
```
```

`chat\_history` Table

| Column Name | Type | Primary Key | Default Value | Notes |
|-------------|----------------------------|-------------|--|------------------------------|
| `id` | `bigint` | Yes | `nextval('chat_history_id_seq')::regclass` | |
| `user_id` | `uuid` | No | | Foreign key to `users` table |
| `role` | `text` | No | | `user` or `model` |
| `message` | `text` | No | | Chat message content |
| `timestamp` | `timestamp with time zone` | No | `now()` | When the message was sent |

```
*  **SQL for `chat_history` table:**
  ```sql
CREATE TABLE chat_history (
 id BIGINT GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
 user_id UUID REFERENCES users(user_id) ON DELETE CASCADE,
 role TEXT NOT NULL,
 message TEXT NOT NULL,
 timestamp TIMESTAMPTZ DEFAULT now()
);
```
```

2. Environment Variables

Ensure the following environment variables are set in your deployment environment (e.g., Vercel, Railway, Render):

- `GEMINI_API_KEY` : Your API key for the Google Gemini API.
- `GLADIA_API_KEY` : Your API key for the Gladia API.

- `FLASK_SECRET_KEY` : A strong, random string used for Flask session management. You can generate one using `os.urandom(24)` in Python.
- `SUPABASE_URL` : Your Supabase Project URL (e.g., `https://abcdefghijklmnopqrstuvwxyz.supabase.co`).
- `SUPABASE_KEY` : Your Supabase `anon` (public) key.

3. Vercel Deployment

1. **Create a Vercel Project:** Link your Git repository (GitHub, GitLab, Bitbucket) containing the application code to a new Vercel project.
2. **Configure Build & Output Settings:**
 - **Framework Preset:** Select `Other` .
 - **Build Command:** `pip install -r requirements.txt`
 - **Output Directory:** `.` (or leave blank)
3. **Environment Variables:** Add the environment variables listed in Section 2 to your Vercel project settings.
4. `vercel.json` : The `vercel.json` file in the root of your project is already configured for Python deployments:

JSON

```
{
  "version": 2,
  "builds": [
    {
      "src": "app.py",
      "use": "@vercel/python"
    }
  ],
  "routes": [
    {
      "src": "/*",
      "dest": "app.py"
    }
  ]
}
```

5. **Deploy:** Vercel will automatically detect the `vercel.json` and deploy your application.

4. Local Development

1. **Clone the Repository:**

Bash

```
git clone <your-repo-url>
cd Health-Hub-Tracker
```

2. Create a Virtual Environment:

Bash

```
python3 -m venv venv
source venv/bin/activate
```

3. Install Dependencies:

Bash

```
pip install -r requirements.txt
```

4. Set Environment Variables:

Create a `.env` file in the root directory and add your API keys and Flask secret key:

Plain Text

```
GEMINI_API_KEY=your_gemini_api_key
GLADIA_API_KEY=your_gladia_api_key
FLASK_SECRET_KEY=your_flask_secret_key
SUPABASE_URL=your_supabase_project_url
SUPABASE_KEY=your_supabase_anon_key
```

Then, load them (e.g., using `python-dotenv` or manually `export` them).

5. Run the Application:

Bash

```
flask run
```

(Or `gunicorn app:app` for a production-like local test)

5. Important Considerations

- **Supabase RLS (Row Level Security):** For a production application, it is highly recommended to implement Supabase Row Level Security to ensure users can only access their own data. This example does not include RLS for simplicity.

- **Error Handling:** Enhance error handling for Supabase interactions to provide more robust feedback.
- **Asynchronous Tasks:** While `ThreadPoolExecutor` helps, for truly scalable background tasks (like long-running Gladia transcriptions), consider dedicated task queues like Celery with a message broker (e.g., Redis) if your hosting platform supports it.